Quest 3: The ISOlation

Lord of the Wings · counting elven votes written in ISO standards

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION
Each vote is a number written in an elven positional system where every digit has its own base; convert every vote to decimal and add them up.

INPUT
One vote per line, such as `ISO-3210 7654`

OUTPUT
The candidate's total score in decimal.

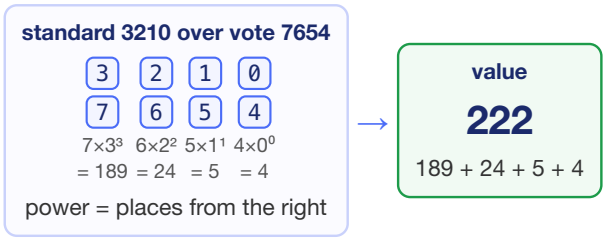
WHAT CHANGES BETWEEN PARTS
II lets the standard be shorter or longer than the vote; III throws out votes that aren't the smallest way of writing their value.

The story

In the Elven Capital, the elves are electing three members of their Grand Council, and the vote count has stalled. Elves don't write numbers in plain decimal. They use many **ISO standards**, each its own positional number system, and all of them count as equally standard. You help convert the votes.

Understanding the question

In decimal every position uses base 10: `7654 = 7×10³ + 6×10² + 5×10¹ + 4×10⁰`. Elves instead write a **standard** that gives **each position its own base**. In `ISO-3210 7654`, the standard `3210` sits over the vote `7654`:



Things to notice

- **`0⁰ = 1`.** A base of 0 in the last position still counts the digit once (the `+ 4` above). The usual power rule already gives 1 here, so no special case is needed.
- **A standard of all one digit is an ordinary base.** `ISO-2222 1011` is binary: 8 + 0 + 2 + 1 = 11. `ISO-88 17` is octal: 1×8 + 7 = 15.
- **Base 1** makes every position worth 1, so `ISO-1111 1111` = 1 + 1 + 1 + 1 = 4.
- **Vote digits can be bigger than their base**, for example 7 in a base-3 position. That is what makes Part III necessary.

PART I Standard and vote have the same length

Worked examples

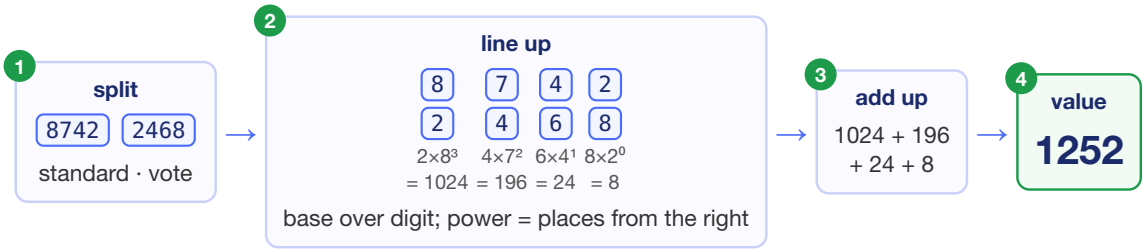
```
ISO-3210 7654   7×3³ + 6×2² + 5×1¹ + 4×0⁰ = 189 + 24 + 5 + 4  =  222
ISO-8742 2468   2×8³ + 4×7² + 6×4¹ + 8×2⁰ = 1024 + 196 + 24 + 8 = 1252
ISO-1111 1111   1×1³ + 1×1² + 1×1¹ + 1×1⁰ = 1 + 1 + 1 + 1      =    4
                                                    total     = 1478
```

Step-by-step solution

1. Remove `ISO-` and split the line into the standard (e.g. `3210`) and the vote (e.g. `7654`).
2. Go through the vote's digits from left to right. For digit number `i`, its power is `length - 1 - i`.
3. Multiply the vote digit by (standard digit) ^ (power) and add it to a running total.
4. That total is the vote's decimal value. Add up every vote's value to get the answer.

How my solution works

Line the standard up over the vote, digit by digit, and add up *digit* × *base*^{power}. `ISO-8742 2468`:



Dots match the step numbers above.

PART II The standard repeats to fit

The standard and the vote can now have different lengths. The standard is a **repeating pattern**: repeat it **from right to left** until it covers the whole vote. The rightmost digits always line up.

Worked examples (sample)

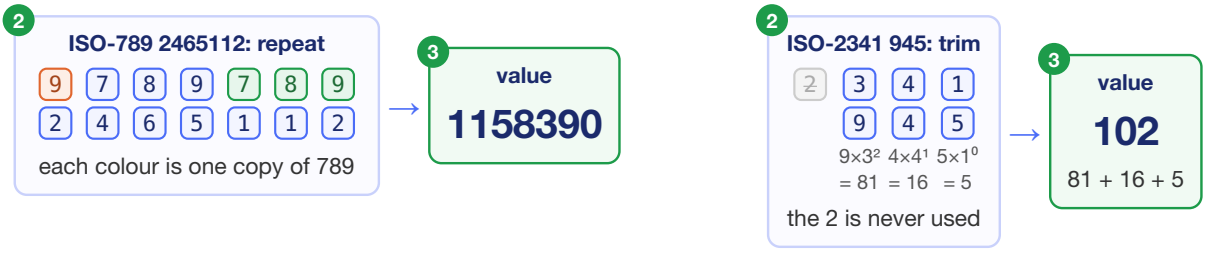
Vote	Expanded standard	Calculation	Value
`ISO-3210 7654`	3210	as Part I	222
`ISO-8742 2468`	8742	as Part I	1252
`ISO-1 1111`	1111	1+1+1+1	4
`ISO-13 777`	313	$7\times3^2 + 7\times1^1 + 7\times3^0 = 63+7+7$	77
`ISO-34 140856`	343434	$1\times3^5 + 4\times4^4 + 0\times3^3 + 8\times4^2 + 5\times3^1 + 6\times4^0$	1416
`ISO-789 2465112`	9789789	$2\times9^6 + 4\times7^5 + 6\times8^4 + 5\times9^3 + 1\times7^2 + 1\times8^1 + 2\times9^0$	1158390
`ISO-2341 945`	341	$9\times3^2 + 4\times4^1 + 5\times1^0 = 81+16+5$	102
Total			**1161463**

Step-by-step solution

1. Split the line into standard and vote, as in Part I.
2. If their lengths differ, build an expanded standard the same length as the vote. Start from the rightmost digit and walk the standard backwards, wrapping around when you run out, putting each digit in front.
3. Compute the value with the expanded standard, exactly as in Part I.
4. Add up every vote's value.

How my solution works

The only new step is **fitting the standard to the vote**: copy it from the right until it is as long as the vote. A long standard simply never has its left-hand digits used.



Dots match the step numbers above.

PART III ISO2, only the smallest representation counts

Because vote digits can be bigger than their base, **one value can be written in several ways**. With standard `5322`, each of these is 137:

```
1028 → 1×5³ + 0×3² + 2×2¹ + 8×2⁰ = 125 + 0 + 4 + 8 = 137
1036 → 125 + 0 + 6 + 6 = 137        (one "2" moved from the last place to the one before)
1044, 1052, 1060, 1103 → also 137
```

ISO2 removes the ambiguity: a vote is **valid only if no smaller number gives the same value** under the same standard. Here only `1028` is valid. Invalid votes count as 0.

The smaller number can have **fewer digits**. In that case the standard is repeated or trimmed to the shorter length, using the same rule as Part II.

Worked examples (sample)

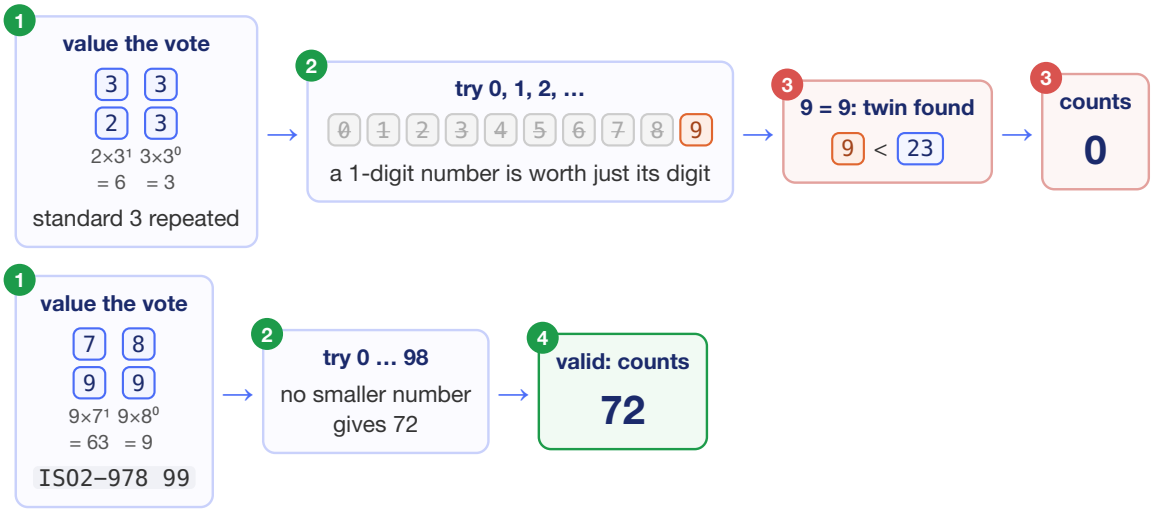
Vote	Value	Smaller twin with the same value	Counted
`ISO2-3 23`	9	`9`	0
`ISO2-61 149`	34	`54`	0
`ISO2-978 99`	72	none	72
`ISO2-5887 5923`	1220	none	1220
`ISO2-76988 123749`	38826	none	38826
`ISO2-435657 929977`	10869	none	10869
`ISO2-3949 870737`	54705	`280207`	0
Total			**50987**

Step-by-step solution

1. Compute the vote's value, exactly as in Part II.
2. For every smaller number 0, 1, 2, … up to vote − 1, compute its value under the **same** standard. The standard is stretched or trimmed to that number's length.
3. If any of them gives the same value, the vote is not the smallest representation: it is invalid and counts as 0. You can stop at the first match.
4. If no smaller number matched, the vote is valid and its value counts.
5. Add up the counted values.

How my solution works

Work out the vote's value, then **try every smaller number** under the same standard (fitted to its own length). The first match makes the vote invalid. To value a candidate it reuses the same calculation with the check switched off, so the check doesn't call itself forever.



Dots match the step numbers above. Red = the smaller twin that makes the vote invalid.

> **Cost:** a valid vote checks every number below it (`ISO2-435657 929977` computes 929,977 values), while an invalid one stops at its first twin (`ISO2-3 23` stops after 10). The puzzle promises "reasonably small" numbers, so brute force is fast enough.

Summary

Part	New rule	What the solution adds
I	each digit has its own base	sum of digit × base^{power}
II	standard repeats from the right	fit the standard to the vote, wrapping from the right
III	only the smallest representation is valid	try every smaller number under the same standard; any match makes the vote 0